Simulation of Squeezed Light Generation in an Optical Resonator

University of Victoria

Aaron Dayton

Supervisor: Prof. Andrew MacRae
Department of Physics and Astronomy
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Why? Quantum Internet

There are existing fiber optic networks all over the world, but current methods for encoding quantum information in optical beams operate at wavelengths not compatible with current infrastructure.

To rapidly deploy a quantum internet across Canada and the world, a cost effective and reliable method for converting optically-encoded continuous-variable quantum information to telecom-compatible wavelengths is required [1].

A potential candidate is hot rubidium-87 gas which may be able to convert information encoded in a near-795 nm beam (which can be produced with modern techniques) to a near-1530 nm beam (which is within the window of the telecom C-band).

The wavelength-conversion happens via optical ***Four Wave Mixing*** and can be resonantly enhanced via ***Cavity Electromagnetically Induced Transparency***.

Four Wave Mixing involves driving an atomic sample by multiple beams to induce the emission of separate beams of ***Squeezed Light*** such that the energy conservation and momentum conservation equations are satisfied.

$$\omega_s + \omega_{II} = \omega_t + \omega_I$$
$$k_s + k_t + k_I + k_{II} = 0$$

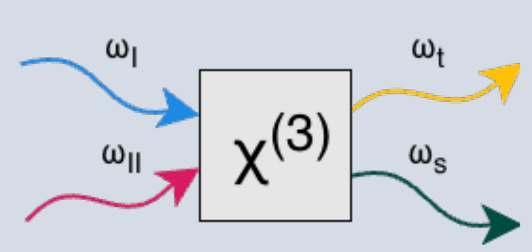

Fig. 1 – Four Wave Mixing Process.

Squeezed Light is with reduced uncertainty in one quadrature (amplitude or phase) but increased uncertainty in the other such that the ***Generalized Uncertainty Principle*** is still observed.

$$\Delta\text{Amplitude} \cdot \Delta\text{Phase} \geq \frac{\hbar}{2}$$

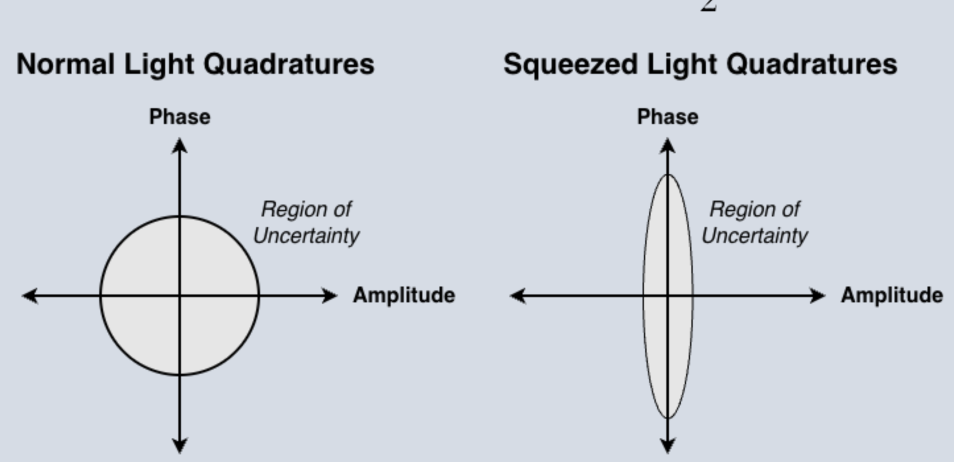

Fig. 2 – Optical Squeezing in amplitude and phase quadratures.

Cavity Electromagnetically Induced Transparency involves driving the atomic sample within an enhancing resonant cavity such that the beams interfere to produce a polarization field which allows some set of frequencies of light to pass through the medium with minimal losses.

How? Setup and Simulation Methods

The overarching goal is to find system parameterizations which induce nonlinear optical processes to create ***Squeezed Light*** quantum correlations*.* We must computationally model the quantum transitions of a hot gas composed of 4-level Diamond-Configuration atoms driven by multiple lasers tuned near-resonance of atomic transitions: two pump lasers stimulating transitions and one signal laser encoded with quantum information.

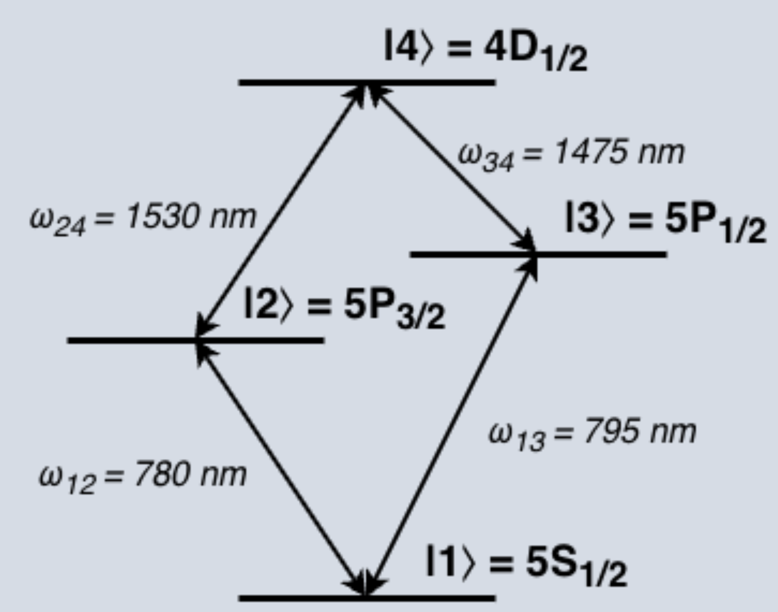

Fig. 3 – Diamond-Configuration Atomic Transition Structure.

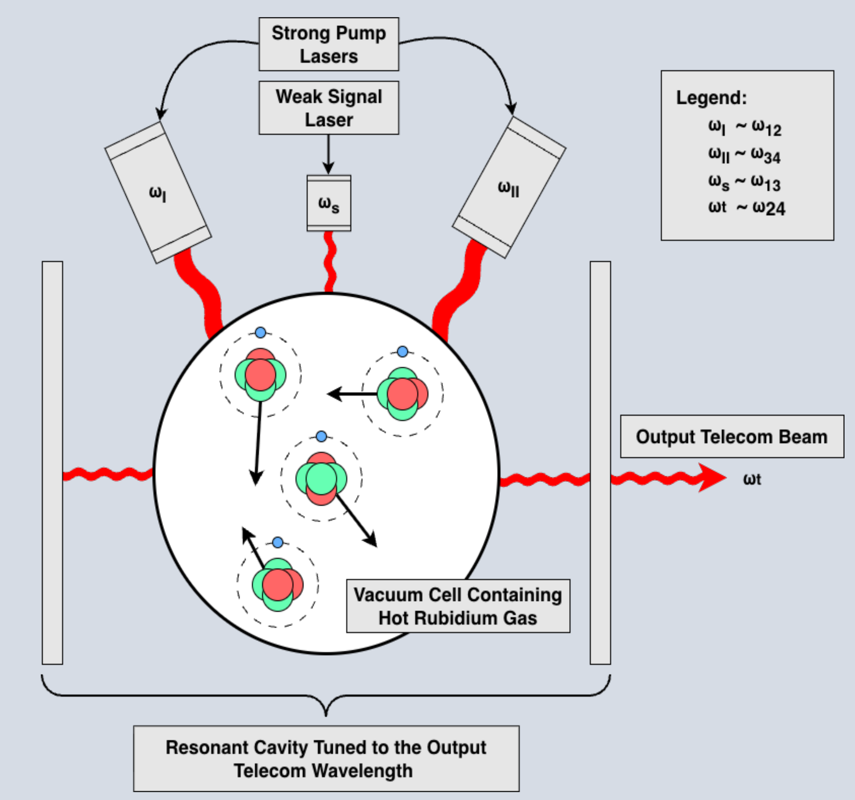

Fig. 4 – Vacuum Chamber Driven within a Resonant Cavity Driven by Lasers Tuned near Transition Frequencies.

To do this we have derived a Hamiltonian $\tilde{H}$ describing the interaction between the atoms and the external fields with atomic velocities $\vec{v}_i$ according to a ***Normal Distribution*** which results in ***Doppler Broadening***.

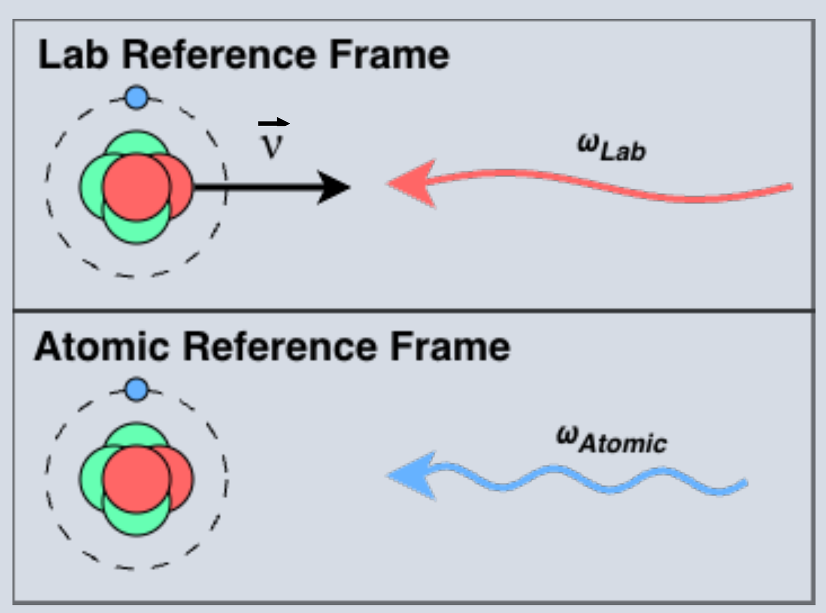

Fig. 5 – Diagram of Doppler Shifting for a single atom.

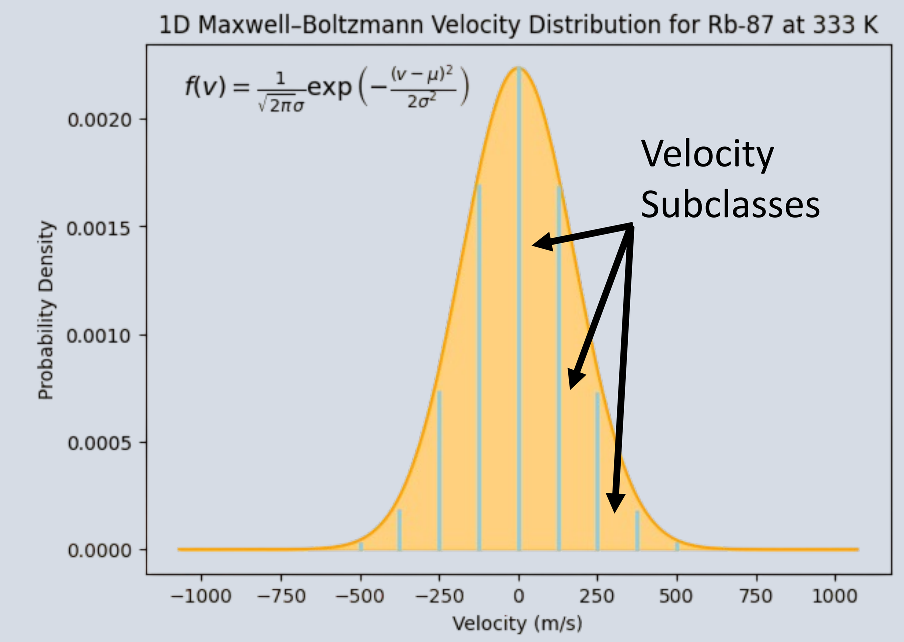

Fig. 6 – Gaussian velocity distribution.

Then the coupled ***Lindblad Master Equation*** and ***Maxwell-Bloch Equations*** are solved iteratively for different velocity subclasses [2] at various ***System Parameterizations*** and the results can be averaged to find when ***Electromagnetically Induced Transparency*** and ***Four Wave Mixing*** phenomena occur simultaneously to generate enhanced ***Squeezed Light***.

Solve for steady state $\hat{\rho}$ and $\hat{d}$

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar}\left[\tilde{H}(\Omega_s, \Omega_t), \hat{\rho}\right] + \sum_i \gamma_i \left(\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \frac{1}{2}\left\{\hat{L}_i^\dagger \hat{L}_i, \hat{\rho}\right\}\right)$$

$$\frac{\partial \Omega_s}{\partial z} = i\frac{\omega_s |d_{13}|^2}{2\epsilon_0 \hbar}\rho_{13} \qquad \frac{\partial \Omega_t}{\partial z} = i\frac{\omega_t |d_{24}|^2}{2\epsilon_0 \hbar}\rho_{24}$$

Solve for Ω_s and Ω_t

This yields theoretical values for the ***Conversion Efficiency*** and ***Squeezing Amplitude*** of the telecom transmission beam.

Then What? Takeaways and Outlook

- With a theoretical value of ***Conversion Efficiency*** and ***Squeezing Amplitude*** either experimental research can be justified or the relatively expensive alternative of using a super-cooled atomic system is supported.
- The ***System Parameterizations*** which yield optimal results can be used as a starting point for experiments.
- If hot atomic ensembles can be used to effectively transmit telecom signals, then connecting nodes in a quantum communication network would be fast, economically viable, and compatible with existing techniques.

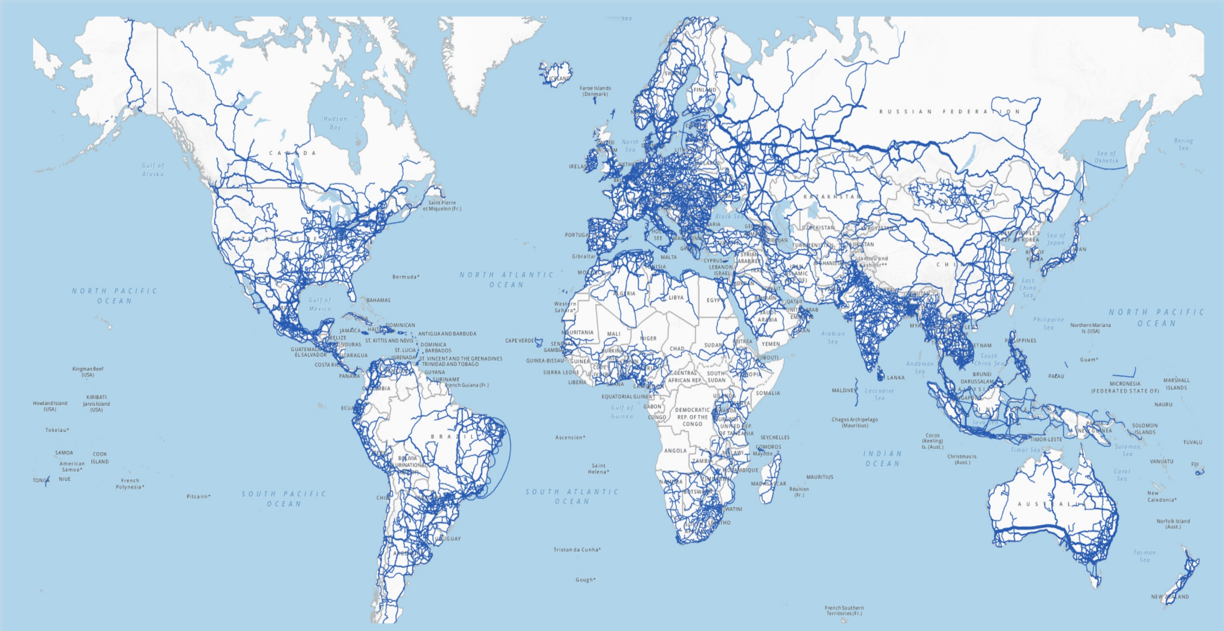

Fig. 7 – Map of current worldwide fiber optic network [3].

Legend

ω_i ≡ Electromagnetic Field Frequency or Transition Frequency
$\hat{\rho}$ ≡ Density Matrix
$\hat{L}_i \equiv i^{th}$ Decay Operator
γ_i ≡ Decay Rate of i^{th} Decay Operator
Ω_i ≡ Electromagnetic Envelope

References

[1]: D. Du, L. Castillo-Veneros, D. Cottrill, G. Cui, P. Stankus, D. Katramatos, J. Martínez-Rincón, & E. Figueroa, "A long-distance quantum-capable internet testbed," Oct. 2024. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2101.12742

[2]: A. MacRae and C. Kupchak, "Propagation Dynamics and transient amplification in warm and cold atomic eit systems," *Optics Communications*, vol. 606, p. 132893, Jun. 2026. doi:10.1016/j.optcom.2026.132893

[3]: International Telecommunication Union - ITU-BDT, "Infrastructure Connectivity Map," ITU, https://bbmaps.itu.int/bbmaps/ (accessed Feb. 24, 2026).